

Multisymplectic observables and higher Courant algebroids

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joint work with Antonio Miti

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Multisymplectic forms

Let M be a manifold, $n \geq 1$.

Definition

An **n -plectic** form is a closed, non-degenerate $\omega \in \Omega^{n+1}(M)$.

Examples:

- i) Volume forms
- ii) $(\wedge^n T^* B, d\theta_{can})$

Next: ω induces

- A) the observables $L_\infty(M, \omega)$
- B) a higher Courant algebroid $E \rightsquigarrow L_\infty(E)_\omega$

A) Observables

Definition (ROGERS 2012)

The L_∞ -algebra $L_\infty(M, \omega)$ of observables is

$$C^\infty(M) \xrightarrow[-n+1]{d} \dots \xrightarrow{d} \Omega^{n-2}(M) \xrightarrow{d} \Omega_{ham}^{n-1}(M, \omega)$$

with higher brackets

$$\{\alpha_1, \dots, \alpha_k\}_k = \pm \iota_{X_{\alpha_k}} \dots \iota_{X_{\alpha_1}} \omega$$

B) Higher Courant algebroids

Consider the “higher Courant algebroid” twisted by ω :

$$(E, \langle \cdot, \cdot \rangle, pr_{TM}, [[\cdot, \cdot]])$$

with

$$E := TM \oplus \wedge^{n-1} T^* M$$

$$\langle \cdot, \cdot \rangle: E \times E \rightarrow \wedge^{n-2} T^* M$$

$$[[(X, \alpha) , (Y, \beta)]] = ([X, Y] , \mathcal{L}_X \beta - \mathcal{L}_Y \alpha - \frac{1}{2} d(\iota_X \beta - \iota_Y \alpha) + \iota_X \iota_Y \omega).$$

Definition (Z. 2012 AFTER FIORENZA-MANETTI 2007, GETZLER 2010)

The L_∞ -algebra $L_\infty(E)_\omega$ is

$$C_{-n+1}^\infty(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^{n-2}(M) \xrightarrow{d} \Gamma_0(E)$$

with certain multibrackets.

The multibrackets involve the Bernoulli numbers B_k :

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} B_k \frac{x^k}{k!}.$$

Main theorem: an embedding of L_∞ -algebras

Theorem (MITI-Z. 2022)

There is an L_∞ -embedding

$$\Psi : L_\infty(M, \omega) \hookrightarrow L_\infty(E)_\omega$$

with components

$$\Psi_1(\xi \oplus \alpha) = \xi \oplus (X_\alpha, \alpha)$$

$$\Psi_k(\xi_1 \oplus \alpha_1, \dots, \xi_k \oplus \alpha_k) = B_{k-1} \sum_{j=1}^k (-1)^{k-j} \iota_{X_{\alpha_1}} \dots \widehat{\iota_{X_{\alpha_j}}} \dots \iota_{X_{\alpha_k}} (\xi_j \oplus \alpha_j)$$

for $k \geq 2$. Here $\xi_i \oplus \alpha_i \in \Omega^{\leq n-2}(M) \oplus \Omega_{ham}^{n-1}(M, \omega)$.

Remark: This was proven

- $n = 2$: by [ROGERS 2013]
- existence for all n : unpublished preprint by [RITTER-SÄMANN 2015]

Remark: Ψ is compatible with gauge transformations by invariant $B \in \Omega^n(M)$.

Sketch of proof

1) Vector space isomorphism

$$\underbrace{L_\infty(M, \omega)}_{\Omega^{\leq n-2}(M) \oplus \Omega_{ham}^{n-1}(M, \omega)} \xrightarrow{\cong} \mathcal{A} \subset \underbrace{L_\infty(E)_\omega}_{\Omega^{\leq n-2}(M) \oplus \Gamma(E)}.$$

Get two $L_\infty[1]$ -algebra structures $(\mathcal{A}[1], m_k)$ and $(\mathcal{A}[1], e_k)$.

Sketch of proof (cont.)

Recall we have an $L_\infty[1]$ -algebra structure $(\mathcal{A}[1], m_k)$.

2) Pushforward:

Any degree 0 linear map

$$p: S^{\geq 1}\mathcal{A}[1] \rightarrow \mathcal{A}[1]$$

↪ degree 0 coderivation C_p of the coalgebra $S^{\geq 1}\mathcal{A}[1]$

↪ automorphism e^{C_p} of $S^{\geq 1}\mathcal{A}[1]$

Get

- a new coderivation $D' := e^{C_p} \circ D_m \circ e^{-C_p}$
- e^{C_p} intertwines D_m and D'

Get

- a new $L_\infty[1]$ -algebra structure m' on $\mathcal{A}[1]$
- an $L_\infty[1]$ -isomorphism from $(\mathcal{A}[1], m_k)$ to $(\mathcal{A}[1], m'_k)$.

Sketch of proof (cont.)

3) Ansatz for p :

As p take the degree 0 linear map

$$\sum_{j=0}^{\infty} c_j \langle \cdot, \cdot \rangle_{-}^j : S^{\geq 1} \mathcal{A}[1] \rightarrow \mathcal{A}[1]$$

where $\langle \cdot, \cdot \rangle_{-}$ corresponds to the skew-symmetric pairing on E .

Find coefficients c_j so that

$$m'_k = e_k.$$

To do so:

Write both in terms of

- m_k ,
- iterated commutators of $\langle \cdot, \cdot \rangle_{-}^j$ with m_1 .

(The commutator bracket on $\text{Coder}(S^{\geq 1} \mathcal{A}[1]) \cong \{S^{\geq 1} \mathcal{A}[1] \rightarrow \mathcal{A}[1]\}$.)

References



E. Getzler

Higher derived brackets

ArXiv:1010.5859



A. Miti and M. Zambon

Observables on multisymplectic manifolds and higher Courant algebroids

ArXiv:2209.05836



C.L. Rogers

L_∞ -algebras from multisymplectic geometry

Lett. Math. Phys. **100** (2012), 29–50



C.L. Rogers

2-Plectic geometry, Courant algebroids, and categorified prequantization

J. Symplectic Geom., **11** (2013), 53–91

Thank you for your attention

Brackets of the L_∞ -algebra $L_\infty(E)_\omega$

- binary bracket:

for $e_i \in \Gamma(E)$ the twisted Courant bracket

$$[e_1, e_2] = [[e_1, e_2]],$$

for $e = (X, \alpha) \in \Gamma(E)$ and $\xi \in \Omega^{\bullet \leq n-2}(M)$

$$[e, \xi] = \frac{1}{2} \mathcal{L}_X \xi.$$

- trinary bracket:

for $e_i \in \Gamma(E)$

$$[e_0, e_1, e_2] = -\frac{1}{6} (\langle [[e_0, e_1]], e_2 \rangle + c.p.).$$

for $\xi \in \Omega^{\bullet \leq n-2}(M)$ and $e_i = (X_i, \alpha_i) \in \Gamma(E)$

$$[\xi, e_1, e_2] = -\frac{1}{6} \left(\frac{1}{2} (\iota_{X_1} \mathcal{L}_{X_2} - \iota_{X_2} \mathcal{L}_{X_1}) + \iota_{[X_1, X_2]} \right) \xi.$$

- n -ary bracket for $n \geq 3$ with n an odd integer:

for $e_i = (X_i, \alpha_i) \in \Gamma(E)$, $[e_0, \dots, e_{n-1}] = \sum_i [X_0, \dots, \alpha_i, \dots, X_{n-1}]$, with

$$[\alpha, X_1, \dots, X_{n-1}] = b_{n-1} \sum_{i < j} (-1)^{i+j+1} \iota_{X_{n-1}} \dots \widehat{\iota_{X_i}} \dots \iota_{X_1} [\alpha, X_i, X_j],$$

for $\xi \in \Omega^{\bullet \leq n-2}(M)$ and $e_i = (X_i, \alpha_i) \in \Gamma(E)$,

$$[\xi, e_1, \dots, e_{n-1}] = b_{n-1} \sum_{i < j} (-1)^{i+j+1} \iota_{X_{n-1}} \dots \widehat{\iota_{X_i}} \dots \iota_{X_1} [\xi, X_i, X_j].$$

Above $b_{n-1} := \frac{(-1)^{\frac{n+1}{2}} 12B_{n-1}}{(n-1)(n-2)}$, where the B_k are the **Bernoulli numbers**:

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} B_k \frac{x^k}{k!}.$$